Finite Elements for Differential Geometry

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March 9, 2026

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Introduction

This lecture on *Finite Elements for Differential Geometry* aims to derive and discuss models, finite elements, and numerical methods for problems arising in (discrete) differential geometry, with a particular focus on applications in continuum mechanics.

The first part is devoted to problems involving extrinsic differential geometry, where the computational domain is embedded in the surrounding Euclidean space, e.g., a plate or surface in 3D. To solve nonlinear shell problems, we first discuss the equations of elasticity (continuum mechanics). It will turn out that this approach has significant weaknesses when one domain dimension is significantly smaller than the others. Thus, we perform dimension reduction, leading to (linear) beams and plates, and reveal several typical discretization problems for those problem classes. Non-standard finite elements like vector-valued tangential-continuous Nédélec, and normal-continuous Raviart–Thomas/Brezzi–Douglas–Marini elements as well as matrix-valued normal-normal-continuous Hellan–Herrmann–Johnson and tangential-tangential continuous Regge elements are motivated, constructed, and analyzed to construct robust discretization methods. Before going to shells, the "prima-donna" of structures, tools from so-called extrinsic differential geometry are repeated, where a surface is embedded in $\mathbb{R}^3$. We introduce the surface's fundamental forms and related quantities (e.g., metric and Weingarten tensor), (covariant) derivatives, and finite elements on surfaces. The discrete (distributional) curvature approximation is discussed. Combining knowledge of differential geometry and finite elements, we derive, discuss, and simulate nonlinear shells that deform under external forces.

In the second part of this lecture, we concentrate on the theory of Riemannian manifolds and the computation and approximation of their intrinsic quantities. Therefore, we do not assume that these manifolds are embedded. In terms of curvature, although the outer normal vector is not available anymore, the Gauss and geodesic curvature can still be computed. The approximation of the metric tensor using solely tangential-tangential continuous Regge finite elements, and the question of computing geodesics and the distributional Gauss curvature, will be fundamental. Again, we use tools of (discrete) differential geometry, such as the Gauss–Bonnet theorem, to define, analyze, and implement the resulting methods.

Hands-on materials and numerical examples can be found on GitHub:

https://github.com/MichaelNeunteufel/2026_FE_DiffGeo

These examples can be run directly in the browser without any installations required.

Chapter 1

Elasticity

This section introduces the equations of continuum mechanics (elasticity) to describe how an elastic body gets deformed in the presence of forces like gravity. For literature on (nonlinear) elasticity and continuum mechanics see e.g. [4, 7, 9, 2, 6].

1.1 Strain tensors

Let $\Omega \subset \mathbb{R}^d$ be an open and bounded domain in $d = 2, 3$ dimensions and let the boundary $\partial\Omega$ be sufficiently smooth. Then $\overline{\Omega}$ describes the reference configuration of a body, also called the undeformed configuration. The boundary is split into a Dirichlet and Neumann part Γ_D and Γ_N, respectively, with $\Gamma_D \cap \Gamma_N = \emptyset$ and $\overline{\Gamma_D} \cup \overline{\Gamma_N} = \partial\Omega$. Further, we assume that the measure of Γ_D is not zero, i.e., $|\Gamma_D| \neq 0$.

Applying external forces to the body leads to *deformation* represented by the function

$$\begin{aligned} \Phi : \overline{\Omega} &\to \mathbb{R}^d \\ x &\mapsto \Phi(x), \end{aligned} \tag{1.1.1}$$

which can be split additively into the identity function and the *displacement* u

$$\Phi = \mathrm{id} + u. \tag{1.1.2}$$

Next, we introduce the *deformation gradient*

$$\boldsymbol{F} := \boldsymbol{I} + \nabla u = \nabla\Phi, \tag{1.1.3}$$

where $\boldsymbol{I}$ denotes the identity matrix. A deformation is called *permissible* if the *deformation determinant* $J := \det(\boldsymbol{F})$ is greater than zero, $J > 0$, which entails that the material is non-interpenetrable, i.e., the orientation is preserved and volume elements with positive measure have also positive measure after the deformation.

To measure the quadratic change of lengths of the deformation, the (right) *Cauchy–Green strain tensor* is introduced as

$$\boldsymbol{C} := \boldsymbol{F}^\top \boldsymbol{F}. \tag{1.1.4}$$

It is also called *metric tensor* in the context of differential geometry and shells. There holds for $\Phi \in C^2(\Omega, \mathbb{R}^d)$ with Taylor's theorem for $\|\Delta x\| \to 0$

$$\begin{aligned} \frac{\|\Phi(x+\Delta x) - \Phi(x)\|^2}{\|\Delta x\|^2} &= \frac{\|\Phi(x) + \nabla\Phi\Delta x + \mathcal{O}(\|\Delta x\|^2) - \Phi(x)\|^2}{\|\Delta x\|^2} = \frac{\Delta x^\top \boldsymbol{F}^\top \boldsymbol{F} \Delta x + \mathcal{O}(\|\Delta x\|^3)}{\|\Delta x\|^2} \\ &= \frac{\Delta x^\top \boldsymbol{C} \Delta x}{\|\Delta x\|^2} + \mathcal{O}(\|\Delta x\|), \end{aligned} \tag{1.1.5}$$

i.e., it measures the change of lengths.

If $\boldsymbol{C} = \boldsymbol{I}$, the body does not get deformed, however, it could be rotated or translated. The following theorem states that these motions are precisely the kernel of $\boldsymbol{C} - \boldsymbol{I}$, so-called *rigid body motions* (or *Euclidean motion*).

Theorem 1.1.1 *Let Ω be a connected domain and $\Phi \in C^1(\Omega, \mathbb{R}^d)$. Then a deformation is a rigid body motion if and only if*

$$\Phi \in RB := \{\Psi(x) = a + \boldsymbol{Q}x \,|\, a \in \mathbb{R}^d,\, \boldsymbol{Q} \in \mathrm{SO}(d)\}, \tag{1.1.6}$$

where $\mathrm{SO}(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, \det \boldsymbol{A} = 1$ *and* $\boldsymbol{A}^{-1} = \boldsymbol{A}^\top\}$ *denotes the special orthogonal group.*

Proof: See e.g., [4, Theorem 1.8-1]. □

This motivates the definition of the *Green strain tensor*

$$\boldsymbol{E} := \frac{1}{2}(\boldsymbol{C} - \boldsymbol{I}) \tag{1.1.7}$$

measuring the real strains induced by the deformation Φ. Its kernel is exactly the rigid body motions, $\ker \boldsymbol{E} = RB$. Inserting (1.1.3)–(1.1.4) yields the representation

$$\boldsymbol{E} = \frac{1}{2}\left(\nabla u^\top \nabla u + \nabla u^\top + \nabla u\right). \tag{1.1.8}$$

Assuming $\nabla u = \mathcal{O}(\varepsilon)$ with $1 \gg \varepsilon > 0$ small and neglecting all higher order terms gives the *linearized strain tensor*

$$\boldsymbol{\varepsilon}(u) := \frac{1}{2}\left(\nabla u^\top + \nabla u\right) = \operatorname{sym}(\nabla u) \tag{1.1.9}$$

used in linear elasticity.

Remark 1.1.2 (Engineering notation) *Objects in reference notation are frequently denoted by capital letters, e.g., X for the position vector in the reference configuration. In contrast, objects in the deformed configuration are denoted by small letters, e.g., $x = \Phi(X)$ for the position vector in the deformed configuration. The deformation gradient is then written as $\boldsymbol{F} = \frac{\partial x}{\partial X}$. Volume elements are related by the deformation determinant $dv = J\,dV$. A line element dx in deformed configuration can be expressed in reference configuration by $dx = \partial x/\partial X\,dX = \boldsymbol{F}\,dX$. So $dx^T dx = dX^T \boldsymbol{C} dX$ and we can define the Green strain tensor by relating the line elements $dx^T dx - dX^T dX = dX^T(\boldsymbol{C} - \boldsymbol{I})dX = 2dX^T\,\boldsymbol{E}dX$.*

1.2 Hyperelastic materials

We consider *hyperelastic constitutive relations*, i.e., the deformation energy is given by a potential $\Psi : \Omega \times \mathbb{R}^{d\times d} \to \mathbb{R}$

$$E_{\text{def}} = \int_\Omega \Psi(x, \boldsymbol{F}(u(x)))\,dx. \tag{1.2.1}$$

More general elastic materials and a derivation based on conservation laws can be found in the literature. Further, for ease of presentation, we assume that the potential is *homogeneous*, i.e.,

$$\Psi(x, \cdot) = \Psi(\cdot). \tag{1.2.2}$$

Given Dirichlet data u_D on Γ_D, a static body load f, and traction forces g on Γ_N we can define the following minimization problem

$$\mathcal{W}(u) := \int_\Omega \Psi(\boldsymbol{F}(u)) - f \cdot u\,dx - \int_{\Gamma_N} g \cdot u\,ds \to \min!. \tag{1.2.3}$$

Gravity is a classical body force, whereas a car driving over a bridge is an example of a traction force.

We seek a function u in the set of admissible displacements given by

$$U := \{u : \Omega \to \mathbb{R}^d \,|\, u = u_D \text{ on } \Gamma_D,\ \det(\boldsymbol{F}(u)) > 0\}. \tag{1.2.4}$$

Note that the constraint $\det(\boldsymbol{F}(u)) > 0$ is often given implicitly in the material law or is neglected in the small deformation case.

To compute the weak formulation of (1.2.3) we take the first variation in direction δu, well-known as the *principle of virtual works*, yielding

$$D_u\mathcal{W}(u)[\delta u] = \int_\Omega \frac{\partial \Psi}{\partial \boldsymbol{F}} : \nabla \delta u - f \cdot \delta u\,dx - \int_{\Gamma_N} g \cdot \delta u\,ds \overset{!}{=} 0 \tag{1.2.5}$$

for all admissible test functions δu, i.e., $\delta u \in U_{\Gamma_D}$, where

$$U_{\Gamma_D} := \{\delta u : \Omega \to \mathbb{R}^d \,|\, \delta u = 0 \text{ on } \Gamma_D\}. \tag{1.2.6}$$

In (1.2.5) we used the Frobenius scalar product of two matrices $\boldsymbol{A} : \boldsymbol{B} = \sum_{i,j=1}^d \boldsymbol{A}_{ij}\boldsymbol{B}_{ij}$ and the chain rule $D_u\Psi(\boldsymbol{F}(u))[\delta v] = \frac{\partial \Psi}{\partial \boldsymbol{F}}(\boldsymbol{F}(u)) : D_u(\boldsymbol{F}(u))[\delta u]$.

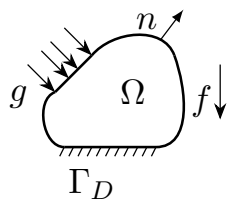

Figure 1.1: Initial configuration of a body Ω with Dirichlet boundary Γ_D, external forces f and g, and outer unit normal vector n.

By defining the *first Piola–Kirchhoff stress tensor*

$$\boldsymbol{P} := \frac{\partial \Psi}{\partial \boldsymbol{F}} \tag{1.2.7}$$

and integration by parts in (1.2.5), we obtain the Euler–Lagrange equation of (1.2.3)

$$\begin{cases} -\mathrm{div}(\boldsymbol{P}) = f & \text{in } \Omega, \\ u = u_D & \text{on } \Gamma_D, \\ \boldsymbol{P}n = g & \text{on } \Gamma_N, \end{cases} \tag{1.2.8}$$

where n denotes the outer normal vector of Ω, cf. Figure 1.1.

Definition 1.2.1 *We call a hyperelastic potential*

- objective *(*frame-indifferent*), if for all positive definite* $\boldsymbol{F} \in \mathbb{M}_+(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, x^\top \boldsymbol{A} x > 0$ *for all* $0 \neq x \in \mathbb{R}^d\}$ *and* $\boldsymbol{Q} \in \mathrm{SO}(d)$

$$\Psi(\boldsymbol{Q}\boldsymbol{F}) = \Psi(\boldsymbol{F}). \tag{1.2.9}$$

- isotropic*, if for all* $\boldsymbol{F} \in \mathbb{M}_+(d)$ *and* $\boldsymbol{Q} \in \mathrm{SO}(d)$

$$\Psi(\boldsymbol{F}\boldsymbol{Q}) = \Psi(\boldsymbol{F}). \tag{1.2.10}$$

For objective materials, its response does not depend on the observer's frame of reference. In other words, the material depends only on stretches, not rotations (all real-life materials are objective). There are no preferred directions for an isotropic material, such as steel. Wood is a classical example of a strongly anisotropic material.

Example 1.2.2 *The potential* $\Psi(\boldsymbol{F}) := \frac{\mu}{4}\left(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I}\right) : \left(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I}\right) + \frac{\lambda}{8}\,\mathrm{tr}(\boldsymbol{F}^\top\boldsymbol{F} - \boldsymbol{I})^2$*, with* $\mu > 0$*,* $\lambda \geq 0$ *is objective and isotropic (Exercise!).*

A crucial consequence of objectivity is that the energy potential only depends on the Cauchy–Green strain tensor.

Theorem 1.2.3 *A potential* $\Psi : \mathbb{R}^{d\times d} \to \mathbb{R}$ *is frame-indifferent if and only if it is a function of the Cauchy–Green strain tensor* $\boldsymbol{C} = \boldsymbol{F}^\top\boldsymbol{F}$*, i.e.,*

$$\Psi(\boldsymbol{F}) = \hat{\Psi}(\boldsymbol{C}). \tag{1.2.11}$$

Proof: See e.g., [2, Chapter VI, Theorem 1.6] or [4, Theorem 3.3-1]. □

We will consider frame-indifferent potentials and write Ψ independently of arguments $\boldsymbol{F}$, $\boldsymbol{C}$, and $\boldsymbol{E}$. Using Theorem 1.2.3, the weak form (1.2.5) can be rewritten as: Find $u \in U$ such that for all $v \in U_{\Gamma_D}$

$$\int_\Omega 2\frac{\partial \Psi}{\partial \boldsymbol{C}}(\boldsymbol{C}(u)) : \mathrm{sym}(\boldsymbol{F}^\top(u)\nabla v)\,dx = \int_\Omega f \cdot v\,dx + \int_{\Gamma_N} g \cdot v\,ds, \tag{1.2.12}$$

where we utilized that $D_u\boldsymbol{C}(u)[v] = D_u\big(\boldsymbol{F}(u)^\top\big)[v]\boldsymbol{F}(u) + \boldsymbol{F}(u)^\top D_u\big(\boldsymbol{F}(u)\big)[v] = 2\,\mathrm{sym}(\boldsymbol{F}^\top(u)\nabla v)$.

Defining the *second Piola–Kirchhoff stress tensor*

$$\boldsymbol{\Sigma} := 2\frac{\partial \Psi}{\partial \boldsymbol{C}} \tag{1.2.13}$$

and exploiting the symmetry of $\boldsymbol{\Sigma}$ gives

$$\int_\Omega \boldsymbol{F}\boldsymbol{\Sigma} : \nabla v\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} g \cdot v\, ds. \tag{1.2.14}$$

Consequently, we obtain the relation

$$\boldsymbol{P} = \boldsymbol{F}\boldsymbol{\Sigma}. \tag{1.2.15}$$

Note that in contrast to $\boldsymbol{\Sigma}$, $\boldsymbol{P}$ is not symmetric. For the sake of completeness, the symmetric *Cauchy stress tensor* $\boldsymbol{\sigma}$ acting on the deformed configuration of the body is introduced by

$$\boldsymbol{\sigma} = \frac{1}{J}\boldsymbol{F}\boldsymbol{\Sigma}\boldsymbol{F}^\top = \frac{1}{J}\boldsymbol{P}\boldsymbol{F}^\top = \frac{1}{J}\frac{\partial \Psi}{\partial \boldsymbol{F}}\boldsymbol{F}^\top. \tag{1.2.16}$$

Then by applying the transformation theorem and the chain rule ($v \circ \Phi^{-1} = \tilde{v}$, $(\nabla_x v) \circ \Phi^{-1} = \nabla_{\tilde{x}}\tilde{v}\boldsymbol{F}$)

$$\int_{\tilde{\Omega}} \frac{1}{J}\boldsymbol{F}\boldsymbol{\Sigma} : (\nabla_{\tilde{x}}\tilde{v}\boldsymbol{F})\, d\tilde{x} = \int_{\tilde{\Omega}} \boldsymbol{\sigma} : \nabla_{\tilde{x}}\tilde{v}\, d\tilde{x}.$$

Assuming a frame-indifferent and isotropic potential yields that the energy potential Ψ is a function of the invariants of the Cauchy–Green strain tensor.

Theorem 1.2.4 (Rivlin–Ericksen) *Let $\mathbb{S}_>(d) := \{\boldsymbol{A} \in \mathbb{R}^{d\times d} \,|\, \boldsymbol{A}^\top = \boldsymbol{A}$ and $\det \boldsymbol{A} > 0\}$ denote the set of symmetric matrices with positive determinant. An energy potential $\Psi : \mathbb{R}^{d\times d} \to \mathbb{R}$ is frame-indifferent and isotropic if and only if there exists a function $\overline{\Psi} : \mathbb{S}_>(d) \to \mathbb{R}$ depending only on the invariants of the characteristic polynomial $\det(\lambda\boldsymbol{I} - \boldsymbol{C}) = \lambda^3 - I_1(\boldsymbol{C})\lambda^2 - I_2(\boldsymbol{C})\lambda - I_3(\boldsymbol{C})$ such that*

$$\Psi(\boldsymbol{F}) = \overline{\Psi}(I_1(\boldsymbol{C}), I_2(\boldsymbol{C}), I_3(\boldsymbol{C})) = \gamma_0(I_1, I_2, I_3)\boldsymbol{I} + \gamma_1(I_1, I_2, I_3)\boldsymbol{C} + \gamma_2(I_1, I_2, I_3)\boldsymbol{C}^2, \tag{1.2.17}$$

where, with $\mathrm{tr}(\boldsymbol{C})$ and $\det(\boldsymbol{C})$ denoting the trace and determinant of $\boldsymbol{C}$,

$$I_1(\boldsymbol{C}) = \mathrm{tr}(\boldsymbol{C}), \qquad I_2(\boldsymbol{C}) = \frac{1}{2}\left(\mathrm{tr}(\boldsymbol{C})^2 - \mathrm{tr}(\boldsymbol{C}^2)\right), \qquad I_3(\boldsymbol{C}) = \det(\boldsymbol{C}). \tag{1.2.18}$$

Proof: See e.g., [8] or [4, Theorem 3.6-1]. □

Example 1.2.5 *From the Rivlin–Ericksen theorem the potential in the Green strain tensor is of the form $\Psi(\boldsymbol{E}) = \Psi(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})$. Assume that Ψ has at $\boldsymbol{E} = 0$ a local minimum with value 0 and is smooth. Defining $\mu := \Psi_{,2}(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})|_{\boldsymbol{E}=0}$ and $\lambda := \Psi_{,11}(\mathrm{tr}(\boldsymbol{E}), \mathrm{tr}(\boldsymbol{E}^2), \det \boldsymbol{E})|_{\boldsymbol{E}=0}$, where $\Psi_{,2}(\cdot,\cdot,\cdot)$ denotes the partial derivative with respect to the second argument and $\Psi_{,11}(\cdot,\cdot,\cdot)$ two derivatives in the first one, there holds*

$$\Psi(\boldsymbol{E}) = \frac{\lambda}{2}\,\mathrm{tr}(\boldsymbol{E})^2 + \mu\,\boldsymbol{E} : \boldsymbol{E} + \mathcal{O}(\|\boldsymbol{E}\|^3).$$

Ignoring the higher-order terms we obtain the *St. Venant–Kirchhoff* material law

$$\Psi_{\mathrm{VK}}(\boldsymbol{E}) = \mu\|\boldsymbol{E}\|_F^2 + \frac{\lambda}{2}\,\mathrm{tr}(\boldsymbol{E})^2, \qquad \|\boldsymbol{A}\|_F^2 := \boldsymbol{A} : \boldsymbol{A}, \tag{1.2.19}$$

is widely used to model nonlinear behavior in a moderate deformation regime. However, it may happen for large deformations that elements get compressed heavily and are even pressed through others. The material law of *Neo–Hooke*

$$\Psi_{\mathrm{NH}}(\boldsymbol{C}) = \frac{\mu}{2}\left(\mathrm{tr}(\boldsymbol{C} - \boldsymbol{I}) - \log(\det(\boldsymbol{C}))\right) + \frac{\lambda}{8}(\log(\det(\boldsymbol{C})))^2 \tag{1.2.20}$$

Figure 1.2: Two solutions of rubber strip clamped on left and right boundary.

prevents the non-physical behavior of the St. Venant–Kirchhoff material law, as there holds

$$\Psi_{\mathrm{NH}}(\boldsymbol{C}) \to \infty \quad \text{for } \det(\boldsymbol{C}) = (\det(\boldsymbol{F}))^2 \to 0 \text{ or } \det(\boldsymbol{C}) \to \infty, \tag{1.2.21}$$

i.e., infinite energy is needed to completely compress or stretch the material. We remark that, especially in the (nearly) incompressible regime, i.e. $\lambda \gg 1$ and thus $\det \boldsymbol{C} \approx 1$, a slightly different material law of Neo–Hooke is also used, namely

$$\tilde{\Psi}_{\mathrm{NH}}(\boldsymbol{C}) = \frac{\mu}{2}\left(\operatorname{tr}(\boldsymbol{C}-\boldsymbol{I}) - \log(\det(\boldsymbol{C}))\right) + \frac{\lambda}{2}(\underbrace{\sqrt{\det(\boldsymbol{C})}}_{=\det \boldsymbol{F}} - 1)^2. \tag{1.2.22}$$

The two material constants $\mu > 0$ and $\lambda > 0$ used above are the so-called Lamé parameters. Two more physically interpretable constants are the Young's modulus $E > 0$, representing the stiffness of a solid material, and the Poisson's ratio $0 \leq \nu < 1/2$[1]. The latter describes the expansion or contraction in the perpendicular direction to the force compressing or stretching the material. Physical units are $[\mu] = [\lambda] = [E] = \frac{N}{m^2} = Pa$ Pascal, whereas the Poisson ratio is dimensionless. To convert these parameters, the following formulae are used

$$\nu = \frac{\lambda}{2(\lambda+\mu)}, \qquad E = \frac{\mu(3\lambda+2\mu)}{\lambda+\mu}, \tag{1.2.23a}$$

$$\lambda = \frac{E\nu}{(1+\nu)(1-2\nu)}, \qquad \mu = \frac{E}{2(1+\nu)}. \tag{1.2.23b}$$

Note that in the limit $\nu \to \frac{1}{2}$, or equivalently $\lambda \to \infty$, the material is called incompressible, necessitating special (numerical) treatment.

1.3 Theoretical results of nonlinear elasticity

The question of existence and uniqueness of this (highly nonlinear) equations is delicate. From physical examples, it is known that we cannot expect uniqueness. Considering e.g., a rubber strip with both ends fixed and neglecting gravity, the identity is a trivial solution for the displacement. When twisting one of the ends by 2π ($360°$), we obtain a different solution to the same boundary conditions, see Figure 1.2. In this manner, we can even produce infinitely many solutions by twisting. Other examples are given, e.g., by buckling (bifurcations) or snap-through problems, where two solutions exist for one external force (one before and one after the snap-through).

For existence, one approach exploits, if possible, the polyconvex structure of the energy potential Ψ in its invariants $I_i(\boldsymbol{C})$, compare (1.2.18), proving the existence of minimizers [1]. The material law of St. Venant–Kirchhoff is not polyconvex, but the Neo–Hooke material law falls in this category. A large class of Ogden-type materials fulfills this assumption. Another ansatz uses the implicit function theorem, obtaining a unique solution for small data. Therefore, relatively strong regularity assumptions must be made [4].

Only a little rigorous mathematical numerical analysis for finite elasticity has been accomplished so far. E.g., in [3] a priori error estimates for finite element discretizations in nonlinear elasticity are discussed for polyconvex materials under the assumption of sufficiently small right-hand sides.

[1] For meta-materials, the Poisson ratio can also take negative values.

1.4 Linearized elasticity

Under the assumption of small deformations all three stress tensors (1.2.7), (1.2.13), and (1.2.16) reduce to one, which we denote by $\boldsymbol{\sigma}$ (e.g., $\boldsymbol{P} = \boldsymbol{F}\boldsymbol{\Sigma} = (\boldsymbol{I} + \nabla u)\boldsymbol{\Sigma} = \boldsymbol{\Sigma} + \mathcal{O}(\varepsilon^2)$ as $\nabla u = \mathcal{O}(\varepsilon)$ by assumption). Assuming a quadratic potential $\Psi(\cdot)$, we deduce a linear stress-strain relation by $\boldsymbol{\sigma} = \frac{\partial \Psi}{\partial \boldsymbol{\varepsilon}} = \mathcal{C}\boldsymbol{\varepsilon}$, where $\mathcal{C}$ denotes the fourth-order elasticity tensor. For an isotropic and frame-indifferent material, the stress-strain relation can be written in Voigt notation as

$$\begin{pmatrix} \boldsymbol{\sigma}_{11} \\ \boldsymbol{\sigma}_{22} \\ \boldsymbol{\sigma}_{33} \\ \boldsymbol{\sigma}_{23} \\ \boldsymbol{\sigma}_{13} \\ \boldsymbol{\sigma}_{12} \end{pmatrix} = \frac{E}{(1+\nu)(1-2\nu)} \begin{pmatrix} 1-\nu & \nu & \nu & & & 0 \\ \nu & 1-\nu & \nu & & & \\ \nu & \nu & 1-\nu & & & \\ & & & 1-2\nu & & \\ & & & & 1-2\nu & \\ & 0 & & & & 1-2\nu \end{pmatrix} \begin{pmatrix} \boldsymbol{\varepsilon}_{11} \\ \boldsymbol{\varepsilon}_{22} \\ \boldsymbol{\varepsilon}_{33} \\ \boldsymbol{\varepsilon}_{23} \\ \boldsymbol{\varepsilon}_{13} \\ \boldsymbol{\varepsilon}_{12} \end{pmatrix}, \tag{1.4.1}$$

or in matrix notation $\boldsymbol{\sigma} = \frac{E}{(1+\nu)(1-2\nu)}((1-2\nu)\boldsymbol{\varepsilon} + \nu \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}) = 2\mu\boldsymbol{\varepsilon} + \lambda \operatorname{tr}(\boldsymbol{\varepsilon})\boldsymbol{I}$. For $\nu \neq \frac{1}{2}$ relation (1.4.1) can be inverted, $\boldsymbol{\varepsilon} = \mathcal{C}^{-1}\boldsymbol{\sigma}$,

$$\begin{pmatrix} \boldsymbol{\varepsilon}_{11} \\ \boldsymbol{\varepsilon}_{22} \\ \boldsymbol{\varepsilon}_{33} \\ \boldsymbol{\varepsilon}_{23} \\ \boldsymbol{\varepsilon}_{13} \\ \boldsymbol{\varepsilon}_{12} \end{pmatrix} = \frac{1}{E} \begin{pmatrix} 1 & -\nu & -\nu & & & \\ -\nu & 1 & -\nu & & & \\ -\nu & -\nu & 1 & & & \\ & & & 1+\nu & & \\ & & & & 1+\nu & \\ & & & & & 1+\nu \end{pmatrix} \begin{pmatrix} \boldsymbol{\sigma}_{11} \\ \boldsymbol{\sigma}_{22} \\ \boldsymbol{\sigma}_{33} \\ \boldsymbol{\sigma}_{23} \\ \boldsymbol{\sigma}_{13} \\ \boldsymbol{\sigma}_{12} \end{pmatrix}, \tag{1.4.2}$$

and $\mathcal{C}^{-1}$ is called the *compliance tensor*. Relation (1.4.2) can be written compactly as

$$\boldsymbol{\varepsilon} = \frac{1}{E}\left((1+\nu)\boldsymbol{\sigma} - \nu \operatorname{tr}(\boldsymbol{\sigma})\boldsymbol{I}\right) = \frac{1}{E}\left((1+\nu)\operatorname{dev}(\boldsymbol{\sigma}) + \frac{1-2\nu}{2}\operatorname{tr}(\boldsymbol{\sigma})\boldsymbol{I}\right), \tag{1.4.3}$$

where $\operatorname{dev}(\boldsymbol{A}) = \boldsymbol{A} - \frac{1}{d}\operatorname{tr}(\boldsymbol{A})\boldsymbol{I}$ denotes the deviatoric (trace-free) part of a matrix. Here, the case $\nu = \frac{1}{2}$ is well-defined leading to the identity $\boldsymbol{\varepsilon} = \frac{1+\nu}{E}\operatorname{dev}(\boldsymbol{\sigma})$, which is unique up to the trace of $\boldsymbol{\sigma}$.

The strong form of (1.2.8) becomes in the linearized case

$$\begin{cases} -\operatorname{div}(\boldsymbol{\sigma}) = f & \text{in } \Omega, \\ u = u_D & \text{on } \Gamma_D, \\ \boldsymbol{\sigma} n = g & \text{on } \Gamma_N, \end{cases} \tag{1.4.4}$$

and the variational problem (inserting the stress-strain relation) reads: Find $u \in U$ such that for all $v \in U_{\Gamma_D}$

$$\int_\Omega \mathcal{C}\boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v)\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} g \cdot v\, ds. \tag{1.4.5}$$

The material laws of St. Venant–Kirchhoff and Neo–Hooke reduce to the linear material law of Hooke

$$\Psi_{\mathrm{H}}(\boldsymbol{\varepsilon}) := \mu\|\boldsymbol{\varepsilon}\|_F^2 + \frac{\lambda}{2}(\operatorname{tr}(\boldsymbol{\varepsilon}))^2 \tag{1.4.6}$$

and we obtain the problem: Find $u \in U$ such that for all test functions $v \in U_{\Gamma_D}$

$$\int_\Omega 2\mu\, \boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v) + \lambda \operatorname{div}(u)\operatorname{div}(v)\, dx = \int_\Omega f \cdot v\, dx + \int_{\Gamma_N} g \cdot v\, ds. \tag{1.4.7}$$

The unique solvability of (1.4.5) is proven in the next section. Note that (1.4.5) and (1.4.7) coincide for $\mathcal{C}$ defined as in (1.4.1).

1.5 Analysis of linear elasticity

We use the classical Lax–Milgram Lemma for elliptic problems to prove existence and uniqueness for the linear elasticity problem (1.4.7).

Lemma 1.5.1 (Lax–Milgram) *Let $a(\cdot,\cdot) : V \times V \to \mathbb{R}$ be a continuous and elliptic bilinear form with constants α and β, respectively. Then, for all $f \in V^*$ there exists a unique solution $u \in V$ of the problem*

$$a(u,v) = f(v), \qquad \text{for all } v \in V \tag{1.5.1}$$

and there holds the stability estimate

$$\|u\|_V \le \frac{1}{\beta}\|f\|_{V^*}. \tag{1.5.2}$$

Surprisingly, the entire gradient of a vector-valued function can be bounded by its symmetric part, cf. [5], at the cost of possibly large constants for anisotropic domains.

Lemma 1.5.2 (Korn's inequality) *Let Ω be a connected and bounded Lipschitz domain (i.e., the boundary can locally be parameterized by a Lipschitz function). Then there exists a constant $\hat{c}_K > 0$ such that*

$$\hat{c}_K^2 \|u\|_{H^1}^2 \le \|u\|_{L^2}^2 + \|\boldsymbol{\varepsilon}(u)\|_{L^2}^2 \tag{1.5.3}$$

for all $u \in H^1(\Omega, \mathbb{R}^d)$ and $\hat{c}_K$ depends only on Ω. Assume further that $\Gamma_D \subset \partial\Omega$ has positive measure. Then there exists a constant $c_K > 0$ such that

$$c_K^2 \|u\|_{H^1}^2 \le \|\boldsymbol{\varepsilon}(u)\|_{L^2}^2 \tag{1.5.4}$$

for all $u \in H^1_{\Gamma_D}(\Omega, \mathbb{R}^d)$. The constants $\hat{c}_K$ and c_K tend to zero for deteriorating aspect ratio.

Korn's inequality relies on the fact that by setting Dirichlet boundary conditions at parts of the boundary, the kernel of the symmetric gradient operator consisting of linearized rigid body motions

$$\ker \boldsymbol{\varepsilon}(\cdot) = RB^{\text{lin}} := \{\Phi(x) = \boldsymbol{A}\,x + b \,|\, b \in \mathbb{R}^d,\ \boldsymbol{A} \in \mathbb{R}^{d\times d} \text{ with } \boldsymbol{A}^\top = -\boldsymbol{A}\} \tag{1.5.5}$$

is locked.

Corollary 1.5.3 *The bilinear form*

$$a(u,v) := \int_\Omega 2\mu\, \boldsymbol{\varepsilon}(u) : \boldsymbol{\varepsilon}(v) + \lambda\, \mathrm{div} u\, \mathrm{div} v\, dx \tag{1.5.6}$$

is continuous and coercive (elliptic) with

$$|a(u,v)| \le (2\mu + \lambda)\, \|u\|_{H^1}\, \|v\|_{H^1}\,,$$
$$a(u,u) \ge 2\mu\, c_K^2\, \|u\|_{H^1}^2\,.$$

Céa's lemma yields the following a-priori finite element error estimate for Lagrange elements $u_h \in U_h \subset U$

$$\|u - u_h\|_{H^1}^2 \le \frac{2\mu + \lambda}{2\mu\, c_K^2} \inf_{v_h \in U_h} \|u - v_h\|_{H^1}^2\,, \tag{1.5.7}$$

from which we deduce that the problem is ill conditioned if the material is nearly incompressible, $\lambda \gg \mu$, or the geometry is anisotropic leading a small Korn constant, $c_K \ll 1$.

Exercise 1.5.4

1. *Proof Corollary 1.5.3 and that the right-hand sides* (1.4.7) *are continuous functionals in $H^1(\Omega, \mathbb{R}^d)$.*
2. *Show that Korn's inequality is sharp by constructing a specific displacement on $\Omega = (0,1) \times (-t/2, t/2)$ and $U = \{u \in H^1(\Omega, \mathbb{R}^2) \,|\, u(0,\cdot) = 0\}$ and showing that $c_K \le c\,t$. Hint: Find u such that $\|\boldsymbol{\varepsilon}(u)\|^2/\|\nabla u\| \le ct^2$.*
3. *Define the pressure-like variable $p = \lambda\, \mathrm{div} u$ as an additional field and rewrite the equations into a Stokes system (with penalty term). Argue that by using a Stokes stable pairing for (u,p), nearly incompressible elasticity can be simulated robustly.*

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